

Newton-Raphson Load Flow Analysis using MATLAB

A modular MATLAB implementation of the **Newton-Raphson Load Flow Analysis** algorithm for a **3-Bus Power System**. The project performs Y-Bus formation, power mismatch calculation, Jacobian matrix construction, line power flow analysis, transmission loss calculation, and voltage profile visualization.

▪ Overview

Load flow analysis is one of the most important studies in power system engineering. This project implements the **Newton-Raphson (NR) method**, an efficient and widely used numerical technique for solving nonlinear power flow equations.

The program reads bus and transmission line data from CSV files, performs iterative load flow calculations until convergence, computes transmission line power flow and system losses, and visualizes bus voltage magnitude and angle profiles.

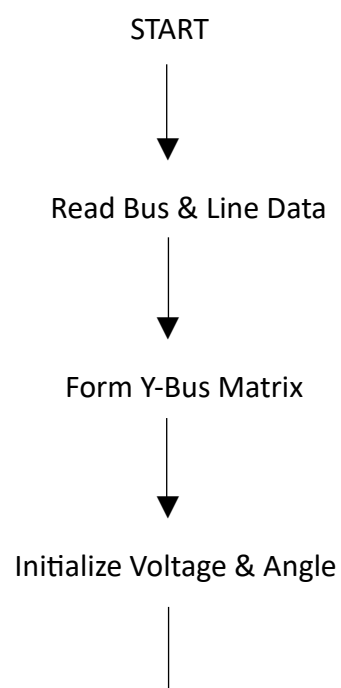
▪ Features

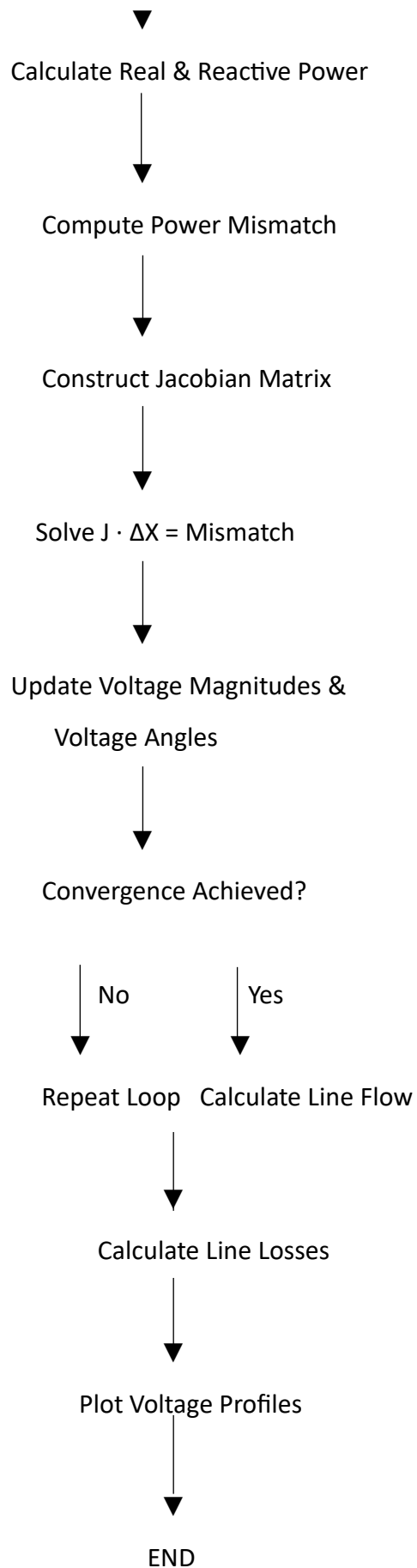
- Reads bus and transmission line data from CSV files
- Forms the Y-Bus Admittance Matrix
- Implements the Newton-Raphson Load Flow algorithm
- Computes real and reactive power mismatches
- Constructs the Jacobian Matrix
- Calculates bus voltage magnitudes and angles
- Computes line currents and complex power flow
- Calculates total real and reactive transmission line losses
- Plots voltage magnitude and voltage angle profiles
- Modular MATLAB implementation using reusable functions

▪ Project Structure

File	Description
main.m	Main program that executes the complete load flow analysis
formYbus.m	Forms the Y-Bus Admittance Matrix
calcPower.m	Calculates real and reactive power at each bus
buildJacobian.m	Constructs the Newton-Raphson Jacobian Matrix
calcLineFlow.m	Calculates transmission line currents and complex power flow
calcLosses.m	Calculates total real and reactive transmission losses
plotVoltage.m	Plots voltage magnitude and voltage angle profiles
BusData.csv	Bus data for the test system
LineData.csv	Transmission line data
README.md	Project documentation

▪ Algorithm Workflow





▪ Requirements

- MATLAB R2021a or later (or any compatible version)
 - Bus data stored in BusData.csv
 - Line data stored in LineData.csv
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▪ How to Run

1. Download or clone this repository.
2. Open the project folder in MATLAB.
3. Ensure all MATLAB files and CSV files are in the same directory.
4. Run:

main

5. View the results in the MATLAB Command Window.
 6. Voltage magnitude and voltage angle plots will be generated automatically.
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▪ Sample Output

The solver successfully converges for the supplied 3-Bus test system.

Load Flow Converged in 4 iterations.

Bus	Voltage(p.u.)	Angle(deg)
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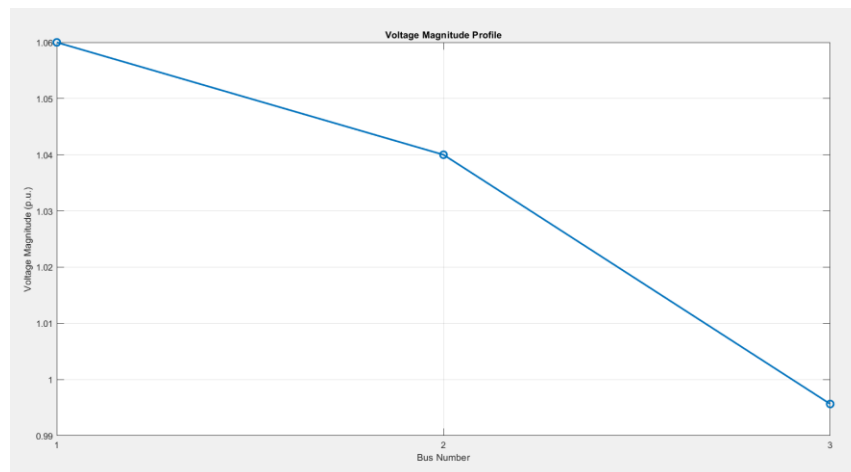
1	1.0600	0.0000
2	1.0400	0.2130
3	0.9956	-2.7029

Results

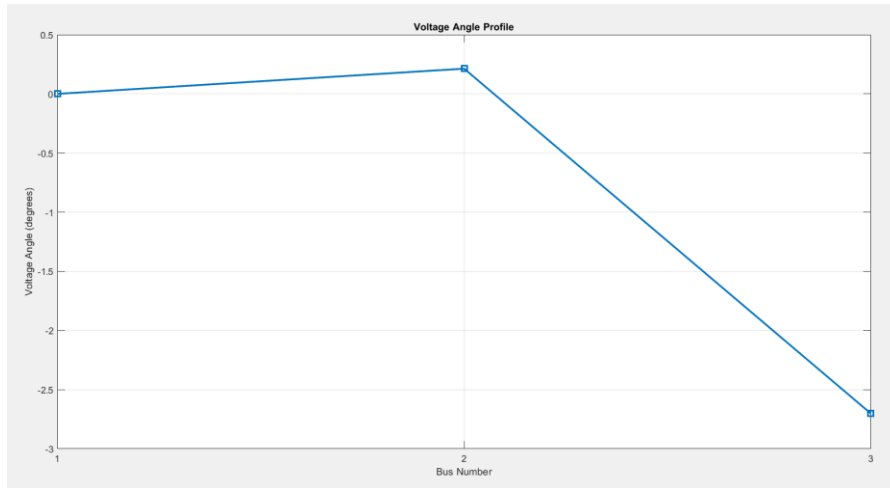
MATLAB Command Window

Command Window	Command Window
<pre>Y-Bus Matrix 6.2500 -18.7500i -5.0000 +15.0000i -1.2500 + 3.7500i -5.0000 +15.0000i 6.6667 -20.0000i -1.6667 + 5.0000i -1.2500 + 3.7500i -1.6667 + 5.0000i 2.9167 - 8.7500i Specified Real Power 0 0.3000 -0.6000 Specified Reactive Power 0 -0.1000 -0.3000 Iteration 1 : Maximum Mismatch = 0.45833333 Iteration 2 : Maximum Mismatch = 0.01058590 Iteration 3 : Maximum Mismatch = 0.00001520 Iteration 4 : Maximum Mismatch = 0.00000000 Load Flow Converged in 4 iterations. Convergence Tolerance : 1.0e-06 ===== NEWTON-RAPHSON LOAD FLOW RESULTS ===== Number of Iterations : 4</pre>	<pre>Number of Iterations : 4 ----- Bus Voltage(p.u.) Angle(deg) ----- 1 1.0600 0.0000 2 1.0400 0.2130 3 0.9956 -2.7029 ----- Load Flow Analysis Completed Successfully. Complex Bus Voltages 1.0600 + 0.0000i 1.0400 + 0.0039i 0.9945 - 0.0470i ===== LINE POWER FLOW RESULTS ===== From To Pij Qij Pji Qji ----- 1 2 0.0446 0.3386 -0.0425 -0.3324 1 3 0.2734 0.1980 -0.2653 -0.1737 2 3 0.3425 0.1496 -0.3347 -0.1263 ----- Total Real Power Loss = 0.017937 p.u. Total Reactive Power Loss = 0.053810 p.u.</pre>

Voltage Magnitude Profile



Voltage Angle Profile



Technologies Used

- MATLAB
- Power System Analysis
- Newton-Raphson Method
- Numerical Methods
- Matrix Computation

Learning Outcomes

This project demonstrates practical implementation of:

- Y-Bus Matrix Formation
- Newton-Raphson Load Flow Algorithm
- Jacobian Matrix Construction
- Power Mismatch Calculation
- Bus Voltage Computation

- Line Current Analysis
 - Complex Power Flow Calculation
 - Transmission Loss Calculation
 - MATLAB Function Modularization
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▪ Future Improvements

- Generalize the solver for any number of buses.
 - Support IEEE 14-Bus, 30-Bus and 57-Bus systems.
 - Include transformer tap-changing and shunt elements.
 - Develop a MATLAB App Designer GUI.
 - Compare Newton-Raphson and Gauss-Seidel methods.
 - Export results to Excel or PDF.
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❖ If you found this project useful, consider giving the repository a star.